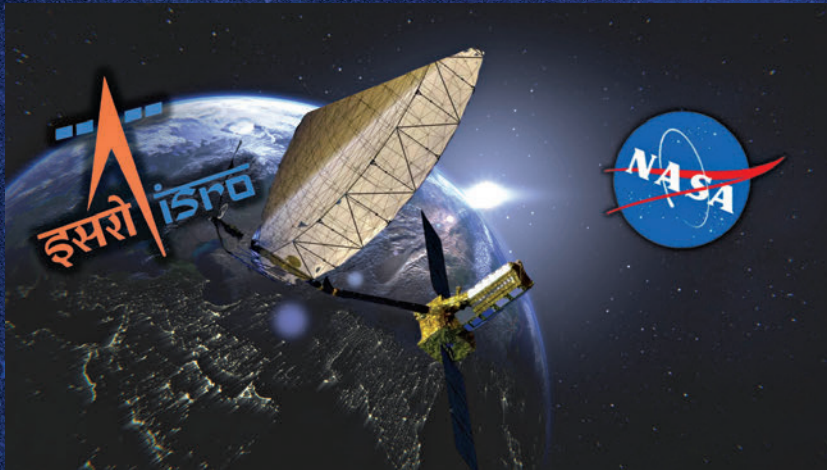


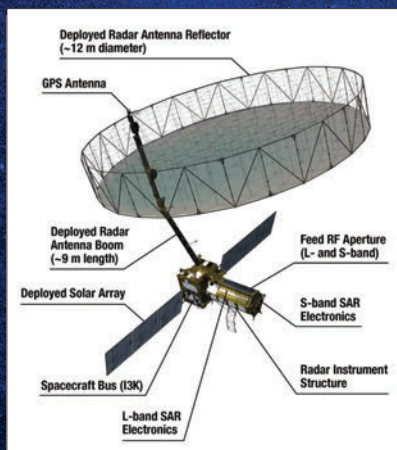
NISAR Earth observation satellite

5 key features explained

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In a significant leap for space-based Earth observation, the NASA-ISRO Synthetic Aperture Radar (NISAR) satellite is officially in orbit. According to ISRO, this Indo-US mission will be one of the most advanced remote sensing systems put into orbit and a game-changer for how we monitor the Earth's surface in real time.



First ever dual-frequency SAR in space

At the heart of NISAR lies its dual-frequency Synthetic Aperture Radar (SAR). Developed through a collaboration between NASA and ISRO, the satellite carries both L-band (1.25GHz) and S-band (3.2GHz) radar systems. They work together using a 12-meter-wide mesh reflector antenna, roughly the size of a tennis court.

Sweeping Coverage with Fine Resolution

NISAR's radar system isn't just about sensitivity, it also brings an unmatched blend of swath width and spatial resolution. The satellite can scan an area 242 kilometers wide in a single pass, while maintaining a resolution range of 3 to 100 meters, depending on the observation mode. This is made possible through an advanced radar technique called SweepSAR, which

balances wide-area scanning with high-resolution imaging.

Rapid global revisit every 12 days

A key strength of NISAR lies in its revisit frequency. Orbiting the Earth at an altitude of 747 kilometers in a sun-synchronous orbit, the satellite will complete a full scan of the planet every 12 days. This consistent revisit cycle is essential for time-sensitive applications, such as tracking how glaciers move, how forests grow or shrink, how agricultural fields develop over seasons, or how coastlines erode. The sun-synchronous orbit ensures that imaging happens under similar lighting conditions each time, which improves the reliability of comparisons over time.

International cooperation, open data for all

NISAR will operate under a fully open data policy, making its satellite imagery and analysis tools freely available to scientists, disaster-response teams, environmentalists, and governments around the world. Most data will be released within 24 to 48 hours of collection, and in some cases, emergency data could be delivered in near real-time. This democratic access to high-quality Earth observation data ensures that everyone can benefit from the insights NISAR provides.

A milestone in global Earth monitoring

It marks a turning point in how humanity observes its own planet, offering unprecedented resolution, frequent global revisit rates, and free, near-real-time data access. NISAR enables breakthroughs in disaster response, climate science, and environmental stewardship. Global anticipation is building each day for the scientific and societal transformation it promises to ignite. **d**